

ROPE HAS A SPECTRAL BUDGET

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ABSTRACT

A RoPE head must place a finite number of rotary frequencies, $K = d_{\text{rot}}/2$, before training begins. We show that this budget is literally a dimension count. Each frequency is a two-dimensional subspace rather than a cosine feature, and block-whitening their pairwise canonical correlations yields the exact identity $r_2 = 2K/(1 + (K-1)\bar{c})$: a table’s usable positional dimension is deflated by the mean redundancy $\bar{c}$ and by nothing else. Geometric RoPE spends heavily on redundancy — at a 4K context, 24 of 64 pairs occupy 48 nominal dimensions worth 2.00 effective ones. Holding K and the sampled endpoints fixed leaves interior allocation as the only remaining handle on $\bar{c}$, and it is a real one: in a 151.9M exact-range matched pair in which only the 30 interior frequencies differ, trained-model OOD NLL moves by $-0.478/-0.205/-0.113$ at 512/1K/2K, with the direction replicated across a 12-configuration, three-seed factorial. Static geometry alone does not predict trained behaviour — substituting tables at inference can raise stable rank from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20 — and we prove that no fixed invertible Q/K map absorbs a table change exactly. The table is a coordinate system fixed before training. As a controlled, zero-parameter intervention on this axis we use EVQ-COSH, the closed-form schedule obtained by quantising the inverse CDF of a convex surrogate’s minimiser. It remains binding at scale: on a 1.485B checkpoint given identical Q/K-only adaptation and identical phase exposure, 8K RULER is 2.02% versus 31.63%, and a matched 8B continuation reaches 14.03% versus 0.295% at 16K.

1 INTRODUCTION

RoPE has a spectral budget. Each attention head receives only $K = d_{\text{rot}}/2$ rotary frequency channels, and those channels must be placed before the model sees a token. Standard geometric RoPE (Su et al., 2024) makes a specific allocation: it spaces the channels uniformly in log-frequency, $\omega_i = b^{-2i/d_{\text{rot}}}$. Because this table is inherited across model families — from LLaMA-3 and Qwen2.5 to DeepSeek-V3 and video DiTs (Grattafiori et al., 2024; Yang et al., 2024; DeepSeek-AI, 2024; Wei et al., 2025) — its allocation is easily mistaken for part of the rotary operator itself.

Recent work makes the distinction consequential. Positional interpolation, YaRN, and LongRoPE-style search transport a realised spectrum toward a target context (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025); FMRoPE retargets the geometric band through its base (Oka et al., 2026); LeRoPE instead learns the individual frequencies and finds stable non-geometric tables (Karypis et al., 2026). Range, transport, and learning all show that the default table is not sacrosanct. They leave a sharper question: *at a fixed spectral range, is the allocation of the finite interior channels an independent training-time variable, and can it be chosen in closed form?*

We answer this in three steps. First we make the budget exact (§3). A rotary pair is a two-dimensional subspace, not a cosine feature, so the right measure of redundancy between two channels is the canonical correlation between their subspaces; block-whitening all K pairs gives the identity $r_2 = 2K/(1 + (K-1)\bar{c})$, in which the nominal budget $2K$ is deflated by the mean redundancy $\bar{c}$ alone. Geometric RoPE spends a great deal on redundancy: at $L=4096$ with $b=5 \times 10^5$ and $K=64$, the 24 slowest pairs occupy 48 nominal dimensions but only 2.00 effective ones. Because $\bar{c}$ is the single quantity allocation can move once K and the endpoints are pinned, this turns *interior allocation* into a well-posed variable rather than a heuristic.

Table 1: Nonexclusive interventions on $\omega_i = b^{-u_i}$. Methods may act at several levels; learned and searched tables differ from EVQ-COSH in how their channel frequencies are obtained, not in whether allocation matters.

	Controlled object	Typical stage	Representative methods
1	deployed phases / spectrum	context extension	PI, YaRN, LongRoPE
2	scalar base b	training or inference	FMRoPE
3	channel frequencies $\{\omega_i\}$	before / during training	EVQ-COSH , LeRoPE, AdaRoPE

Second, we test whether that variable is real for a trained model. In an exact-range identification experiment (§4.2) two arms share the sampled highest and lowest frequencies, log-span, initialisation, token order, optimiser, training budget, and evaluation anchors; only the $K-2 = 30$ interior frequencies move. After training, endpoint-normalised cosh minus the uniform-in-log grid changes OOD NLL by $-0.478 / -0.205 / -0.113$ at 512/1K/2K in the seed-42 matched pair, and a separate 12-configuration, three-seed factorial replicates the direction in 10/12 and 9/12 configurations for two structurally different non-uniform schedules.

Third, we bound what the geometry explains. Static identifiability is necessary but not sufficient: in a frozen 50M diagnostic, substituting the EVQ-COSH table at inference raises stable rank from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20, and the $\text{table} \times \text{weights}$ interaction on LM loss is $5.9 \times$ the size of either main effect. We prove that this is structural — no fixed, position-independent invertible Q/K map can absorb a nontrivial table change exactly — so a table is a coordinate system fixed before training, and the weights learn coefficients in it.

Against this backdrop EVQ-COSH is a construction, not the theory’s unique answer. Representing a table by a density over log-frequency and minimising a convex collision surrogate gives a cosh density whose inverse CDF quantises to a closed-form warp of the RoPE inverse-frequency initialisation (§3.4). It adds no learned parameters, auxiliary loss, or search, and recovers geometric RoPE as $\tau \rightarrow 0$; we use it as a reproducible, zero-parameter intervention on the allocation axis.

The effect survives beyond mechanism-scale controls (§4.3). On OLMo-2 (1.485B), Native and EVQ-COSH receive the same 300-step Q/K-only continuation, data, module budget, and 4K/8K/16K phase exposure; the active method variable is the frequency table. At 8K, Native/EVQ-COSH scores are 2.02/31.63% on RULER and 0/17.5% exact on held-out 2WikiMultiHopQA. Matching long-phase exposure therefore does not erase the frequency-substrate effect. On LLaMA-3-8B, a separate matched physical-8K continuation moves 16K RULER from 0.295% to 14.03%.

2 RELATED WORK

Write the RoPE table as $\omega_i = b^{-u_i}$, with the geometric grid $u_i = 2i/d_{\text{rot}}$. Prior work changes several, nonexclusive parts of this parameterization: it transports a pretrained spectrum to a longer range, changes the scalar base, or changes the frequencies assigned to individual channels (Table 1). This decomposition separates when a method acts from which spectral degrees of freedom it controls.

Context-range extension. Relative position information can be injected additively, through learned or biased attention terms (Shaw et al., 2018; Press et al., 2022; Chi et al., 2022), or multiplicatively through rotation (Su et al., 2024; Sun et al., 2023); we take the rotary operator as given and ask only where its frequencies are placed. Position interpolation rescales positions to keep rotary phases within the training range (Chen et al., 2023). YaRN, SelfExtend, PoSE, and CLEX refine this extension through frequency-dependent scaling, remapped positions, or extension-time training (Peng et al., 2024; Jin et al., 2024; Zhu et al., 2024; Chen et al., 2024). LongRoPE and LongRoPE2 search channel-wise rescaling factors for pretrained models (Ding et al., 2024; Shang et al., 2025). These methods may combine several intervention levels; their common goal is to transport a learned model to a declared longer context. Bounds based on the RoPE base characterize limits of this regime (Xu et al., 2024).

Base frequency and emergent bands. RoPE models use the geometric spectrum non-uniformly. Barbero et al. identify frequency-localized structure in query and key norms and introduce partial RoPE, which removes the slowest rotations (Barbero et al., 2025). Oka et al. show that the high-norm band’s location depends jointly on the scalar base and the training length (Oka et al., 2026). Their FMRoPE rule sets $b = L_{\text{train}}$ during training or fine-tuning and may retarget b to the evaluated length at inference. Thus FMRoPE moves the geometric spectrum by changing one scalar while retaining a uniform exponent grid. By contrast, an exact-range allocation test can hold the endpoints and log-frequency span fixed while changing only the interior grid (§4.2).

Learning and constructing frequency tables. LeRoPE learns one log-scale per frequency band, shared across layers and heads, and reports consistent non-geometric profiles on a 52M–2.5B model ladder (Karypis et al., 2026). In its 217M ablation, retraining with a frequency table frozen from an independent LeRoPE run retains 63.6% of the full validation-perplexity gain over RoPE, compared with 10.4% for partial RoPE. This isolates substantial value in the fixed table, although obtaining that table still requires a separate learned run. AdaRoPE instead learns per-head frequencies together with head-specific, length-dependent attention scaling (Wang et al., 2026). Recent analysis further shows that the frequencies used by a trained model depend on the relative-distance structure of its training data, with the preferred frequency varying inversely with dependency width (Wu et al., 2026).

Resonance RoPE analytically snaps each wavelength to a nearby integer period to close phase gaps during interpolation, producing a fixed non-geometric table that can be combined with range scaling (Wang et al., 2024). EVQ-COSH instead derives an interior allocation density from a variational objective and tests that variable with the sampled range held fixed.

EVQ-Cosh. EVQ-COSH addresses a different fixed-table objective: it warps the exponent grid before training using the closed-form solution of a stated variational surrogate. The result is shared across heads, introduces no learned positional parameters, and does not require a deployment target. It therefore complements range extension and differs from LeRoPE and AdaRoPE in how the non-geometric table is obtained, not in whether frequency allocation matters. Alternatives that learn a broader positional function, including DAPE, FIRE, and FoPE (Zheng et al., 2024; Li et al., 2024; Hua et al., 2025), change a larger functional object than a fixed RoPE table.

Evaluation. Long-context evaluation spans retrieval diagnostics and task suites (Bai et al., 2024; Hsieh et al., 2024), and information can be lost despite being present in the prompt (Liu et al., 2024; Zhang et al., 2024). We use PE-dominant diagnostics for mechanism and RULER for effective-context claims, keeping task-family adaptation distinct from unseen-task transfer.

3 THEORY

Figure 1 summarises the construction. We keep three layers apart, because they carry different epistemic weight: *exact* statements about the geometry a finite table can express (§3.1–§3.3); a *modelling choice* that turns one of them into a closed-form schedule (§3.4); and an *empirical* operating point. No experimental claim in §4 rests on the first layer alone — indeed §3.3 shows it cannot.

3.1 A FINITE TABLE IS A SET OF TWO-DIMENSIONAL SUBSPACES

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}} = d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

A rotary pair contributes $f_{\omega}(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_{\omega} = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (1)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_{ω} and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_{\omega}(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_{\omega} = \mathbb{E}[x_{\omega}^{\top} x_{\omega}]$ and $H_{\omega\nu} = \mathbb{E}[x_{\omega}^{\top} x_{\nu}]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_{\omega}^{-1/2} H_{\omega\nu} S_{\nu}^{-1/2}$, the singular values of $Q_{\omega\nu}$ are

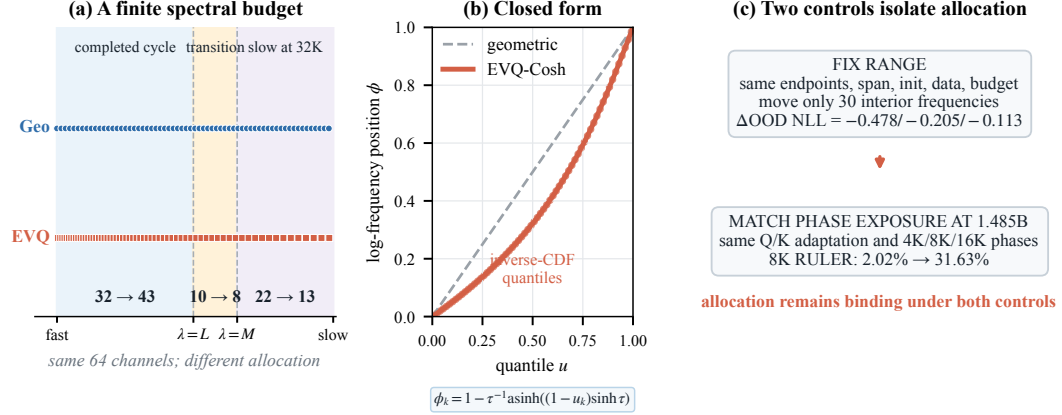


Figure 1: **EVQ-COSH frequency allocation.** Geometric RoPE spaces channels uniformly in log-frequency; EVQ-COSH applies a closed-form inverse-CDF warp that reallocates the same *number* of channels toward wavelengths completing a training-window cycle. Rotation, architecture, and optimiser are unchanged.

the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (2)$$

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. A).

3.2 THE SPECTRAL BUDGET IS A DIMENSION COUNT

Block-whitening every pair gives a global correlation Gram R with diagonal blocks I_2 . Write $\bar{c}$ for the mean of the c_{ij} .

Theorem 1 (Spectral budget identity). $\text{tr } R = 2K$ and $\text{tr}(R^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 (stable) effective dimension of the positional basis is exactly

$$r_2(R) = \frac{(\text{tr } R)^2}{\text{tr}(R^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (3)$$

The nominal budget $2K$ is deflated by the average pairwise redundancy, and by nothing else. This is the sense in which RoPE has a spectral budget, and it names the variable a schedule acts on: *once K and the sampled endpoints are fixed, allocation moves exactly one quantity, $\bar{c}$.* The exact-range experiment of §4.2 manipulates precisely that. Note also what (3) does not do: it is symmetric in the frequencies and cannot single out a particular density shape — a prediction §4.4 confirms.

The deflation is concentrated at the slow end.

Proposition 2 (Low-frequency collapse). As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$, and for slow channels $x = \omega L$, $y = \nu L$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric $F = \text{diag}(p) - pp^\top$, which annihilates the constant direction, the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.

Concretely, at $L=4096$, $b=5 \times 10^5$, $K=64$ the 24 pairs with $\omega L \leq 1$ occupy 48 nominal dimensions but have block-whitened $r_2 = 2.00$: about 96% of that sub-budget is redundant. Slow channels are *redundant*, not unused — they still carry content, and Prop. 2 says only that many of them describe nearly the same positional direction.

3.3 THE TABLE IS A TRAINING-TIME COORDINATE SYSTEM

Theorem 1 constrains what a table can express. It does not predict what a trained model does with one, and the gap is not small. In a frozen 50M diagnostic ($L=512$, $b=5 \times 10^5$; no training, no

Table 2: **Static geometry does not predict trained behaviour.** Two 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations; no training, no GPU). The worst cell is also the one whose static stable rank improves most.

Trained weights	Runtime table	PPL	Stable rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

gradient) we cross two trained weight sets with two runtime tables (Table 2). Substituting EVQ-COSH for the geometric table at inference *raises* the block-whitened stable rank from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20.

Two consequences discipline the rest of the paper. First, no static criterion — collision, stable rank, or log-determinant — may serve as a proxy for trained quality, so every task claim in §4 is made on trained models. Second, the two *self-consistent* systems are essentially tied in-window (7.14 vs. 7.16) while both mismatched cells collapse: on LM loss the table and weight main effects are +0.599 and −0.597, and their interaction is −3.537, about $5.9\times$ larger (configuration-level bootstrap CI $[-5.17, -3.04]$). The table behaves as a coordinate system fixed before training, with the weights learning coefficients in it.

This is not a tuning failure that a better adapter would remove.

Theorem 3 (Post-hoc transplant obstruction). *Let $R_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top R_{\Omega'}(\Delta) B = R_\Omega(\Delta)$ for all Δ in a set containing an open interval, then Ω' and Ω have the same frequency multiset, up to sign, permutation, and change of basis inside a pair.*

The proof is short: $\Delta=0$ gives $A^\top B = I$, so the relation becomes a similarity $A^\top R_{\Omega'} A^{-\top} = R_\Omega$; differentiating at zero makes the generators similar, and similarity preserves the spectrum $\{\pm i\omega_k\}$ (App. A). A nontrivial table change therefore cannot be absorbed exactly by any fixed linear Q/K compensation, and the in-window cost of retrofitting a pretrained checkpoint is structural rather than an artefact of insufficient adaptation. The theorem does not rule out approximate repair by retraining, nonlinear maps, or the V/O and residual paths — which is what the adaptation protocols of §4.3 do.

3.4 A CLOSED-FORM ALLOCATION

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a discrete allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. Choosing ρ needs a tractable proxy for $\bar{c}$. The exact kernel is oscillatory and is not pointwise approximated by a smooth two-term model, so we analyse the convex surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (4)$$

with $\alpha, \beta > 0$ fitted to the exact kernel on a finite channel grid (App. A.14). The diagonal term penalises channel-load concentration; $\min(\phi, \psi)$ is the Green kernel of $-\partial_\phi^2$ and penalises pairs of channels that are *both* slow — the regime Prop. 2 identifies as redundant.

Theorem 4 (Stationary allocation under the surrogate). *Under (4) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (5)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Corollary 5 (Closed-form quantiles). *$F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so*

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (6)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

The full stationary density also admits a Fisher-forced particular solution; we deploy the homogeneous *pure-tether* branch (5) because it alone has an analytically invertible CDF (App. A.4).

The recipe, and the status of τ . With d_{eff} the effective head dimension, equal to d_{head} for the full-rotary architectures studied here,

$$\tau = d_{\text{eff}}/\sqrt{L_{\text{train}}}, \quad u_k = (k + \frac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (7)$$

Set τ , evaluate (6), replace the inverse frequencies: no auxiliary loss, no extra parameters, no declared deployment length. A post-softmax transport model supplies the $d_{\text{eff}}/\sqrt{L_{\text{train}}}$ form but not its constant, and its small- τ expansion does not certify the deployed $\tau \approx 4$ regime (App. A.17); we therefore treat $c=1$ as a zero-search default, and across the 21 configurations of §4.4 the selected τ never leaves a $0.75 \times -1.5 \times$ window around it. Finally, two grids appear below and are *not* interchangeable: EVQ-COSH deploys the midpoint grid of (6), while the endpoint-normalised variant $u_k=k/K$ ($\phi_0=0$ preserves $\omega_0=1$) is a shape diagnostic used where a control must pin the sampled extrema exactly, as in §4.2.

3.5 WHERE THE FINITE BUDGET MOVES

In wavelength $\lambda(\phi) = 2\pi b^\phi$, a training span L and evaluation ceiling $M > L$ split the axis into three regions: $\lambda \leq L$ completes at least one cycle during training; $L < \lambda \leq M$ varies across the target window without completing one; $\lambda > M$ completes none even at the ceiling.

Lemma 6 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at $\phi_c(\tau) = 1 - \tau^{-1} \text{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3} \approx 0.4226$.*

For OLMo-2 ($b=5 \times 10^5$, $L=4096$, $M=32768$) the boundaries $\phi_L=0.494$ and $\phi_M=0.652$ lie to the right of that universal bound, so at $\tau=2$ the 64-pair table moves from 32/10/22 channels to 43/8/13: eleven channels bought for the completed-cycle region by spending density in both longer-wavelength regions. This is an exact budget statement, not a trained-loss theorem, and it yields the falsifiable prediction tested in §5. Proof and a divergence inequality are in App. A.11.

4 EXPERIMENTS

4.1 SETUP AND ENDPOINTS

NLL/PPL, teacher-forced NLL-gap retrieval, causal source use, strict autoregressive exact match, task F1/exact match, and RULER macro are kept as separate endpoints throughout; EVQ-COSH changes only the training-time frequency table. Controlled from-scratch runs use fixed GPT-style decoder stacks (Radford et al., 2019); corpora, hyperparameters and seed scope are in App. H and with each result.

Range-scaling arm. Arms labelled +RAMP use our fixed-index smooth-ramp scaler from the NTK-by-parts / YaRN family (Peng et al., 2024), applied identically to Geo and EVQ-COSH so that only the substrate differs. It is not an official YaRN reproduction or a jointly tuned range-method comparison (App. D).

4.2 IDENTIFYING THE ALLOCATION AXIS

Protocol. In the seed-42 pair, two 151.9M models share architecture, trainable initialisation, token order, optimiser, LR schedule, batch geometry, a 499,974,144-token budget, and 32 frozen evaluation anchors. The baseline is our paper-faithful implementation of the uniform-in-log rule in Oka et al. (2026, §6.1), using training base 256; the intervention is endpoint-normalised cosh spacing at $\tau=4$. Their sampled extrema and log-frequency span are identical, so only the $K-2 = 30$ interior frequencies move. We measure paired final-128-token teacher-forced NLL.

Result. At the fixed training range, cosh-minus-uniform NLL is $-0.478 / -0.205 / -0.113$ at 512/1K/2K in the seed-42 arm (Fig. 2a). Since every sampled scalar range quantity is pinned, the effect cannot be attributed to base or span selection: *fixed range + different interior allocation $\Rightarrow$ different trained-model behaviour.*

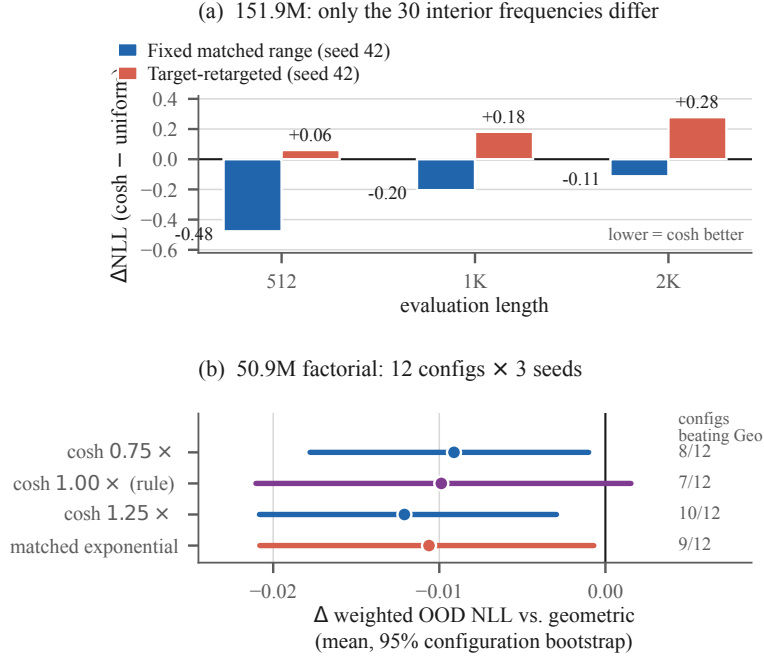


Figure 2: **Allocation is identifiable.** (a) With sampled extrema and log-span pinned, moving only the 30 interior frequencies shifts trained-model OOD NLL (blue, seed 42; anchors 32/32, 27/32, 22/32); the contrast reverses once *both* grids are target-retargeted (red; 5/32, 1/32, 2/32). (b) In a separate three-seed, 12-configuration factorial the 1.25 $\times$ Cosh and matched-exponential arms favour non-uniform allocation in 10/12 and 9/12 configurations.

Table 3: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times 4 \times 8$, paired within configuration and seed. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
Cosh 0.75 $\times$ – Geo	-0.00912	[-0.018, -0.001]	8/12	0.082
Cosh rule – Geo	-0.00988	[-0.021, 0.002]	7/12	0.125
Cosh 1.25 $\times$ – Geo	-0.01210	[-0.021, -0.003]	10/12	0.027
Matched exponential – Geo	-0.01062	[-0.021, -0.001]	9/12	0.071
<i>Cosh and the matched exponential are not empirically separated:</i>				
Cosh (rule) – exponential	+0.00074	[-0.006, 0.008]	7/12	0.836

Cross-configuration replication. A pre-specified 50.9M factorial spans two bases, two training lengths, three head dimensions, and three paired seeds, with additional boundary- τ arms. It repeats the exact-range control and scores weighted teacher-forced OOD NLL over $2 \times 4 \times 8$, using the 12 structural configurations as the statistical units (Table 3). The pre-specified 1.25 $\times$ cosh and matched-exponential schedules favour the non-uniform allocation in 10/12 and 9/12 configurations. The factorial establishes cross-configuration direction; the 151.9M pair provides the direct effect size.

Range and allocation are different controls. Retargeting *both* grids to each evaluation length reverses the seed-42 ordering (Fig. 2a, red). Fixed-range allocation and target-aware range selection therefore answer different questions: the former identifies the interior-allocation axis; the latter can dominate once range is allowed to track the target.

Table 4: **Mature-scale effective-context results.** Columns are relative to each model’s physical training cap (4K for OLMo-2, 8K for LLaMA-3-8B). Arms within a protocol receive identical adaptation and phase exposure; RULER is official macro, 2Wiki is greedy normalized exact. The parenthetical NIAH figure is an independent EVQ-COSH training seed (App. F).

Model / protocol	Endpoint, arm	1×	2×	4×
OLMo-2 1.485B, sel. Q/K	RULER: Native	72.19%	2.02%	0.38%
	RULER: EVQ-COSH	42.44%	31.63%	5.03%
	2Wiki exact: Native	22.0%	0%	0%
	2Wiki exact: EVQ-COSH	21.5%	17.5%	4.0%
OLMo-2 1.485B, Q/K/V/O	NIAH first-num. exact: Native	—	0/100	—
	NIAH first-num. exact: EVQ-COSH	—	69/100 (67/100)	—
LLaMA-3-8B, Q/K/V/O	RULER: Native	94.44%	0.295%	—
	RULER: EVQ-COSH	77.60%	14.03%	—

4.3 PHASE-MATCHED ATTRIBUTION AT MATURE SCALE

1.485B: matching phase exposure does not erase the gap. On OLMo-2-0425-1B-Instruct (OLMo Team, 2025), Native and EVQ-COSH start from their matched Stage-A parents, freeze inherited V/O LoRA tensors, and continue only Q/K LoRA for 300 steps. Both arms use the same data, order, optimiser, trainable parameter count, and 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure; physical sequences remain at most 4K. The active method variable is the fixed frequency table. On all 13 RULER families (Hsieh et al., 2024), Native/EVQ-COSH official macro is 72.19/42.44% at 4K, 2.02/31.63% at 8K, and 0.38/5.03% at 16K. Native receives the same long-phase exposure yet remains at 2.02% at 8K: exposure alone cannot explain the 15.7× score gap.

1.485B held-out multi-hop QA. The same Q/K-only design, trained independently for 2Wiki-MultiHopQA, gives Native/EVQ-COSH exact match of 22.0/21.5% at 4K, 0/17.5% at 8K, and 0/4.0% at 16K on 200 held-out examples per length (Bai et al., 2024); token F1 is 25.99/24.84%, 0.07/21.48%, and 0/8.57%. The two substrates acquire nearly identical in-window QA while only EVQ-COSH carries that capability to the longer positions.

Independent 1.485B generation and language modelling. A matched rank-64 Q/K/V/O continuation (Hu et al., 2022) moves 8K `niah_single_1` strict first-generated-number exact from 0/100 to 69/100, with 67/100 for an independently trained EVQ-COSH seed; on a disjoint set whose source-to-generation gaps all exceed the largest training gap the scores are 0/100 versus 49/100 and 48/100, and natural-text NLL moves from 3.735/4.851 to 2.703/2.925 at 8/16K. A same-initialisation, same-recipe step-0 → 1,000 trajectory over 2.097B tokens — not a bitwise paired run, since the two arms use different trainer stacks — gives EVQ-COSH-minus-Geo NLL of +0.072/ +0.038/ −0.044/ −0.135 at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8/16K.

8B causal source use and RULER. On LLaMA-3-8B-Instruct (Grattafiori et al., 2024), matched seed-42 ordinary-LM adaptation gives EVQ-COSH-minus-Native-LoRA NLL of +0.390/ −1.510/ −2.048 at 8/16/32K. On ten frozen true-16K retrieval cases, median target-block hit@16 over 32 heads rises from 18.75% to 64.06%, and deleting the remote gold block changes NLL by −0.0095 for Native against +1.5055 for EVQ-COSH: causal source use, not merely attention mass. A separate identical physical-8K, 13-family adaptation yields official macro 94.44/77.60% at 8K and 0.295/14.03% at 16K; normalized exact is 17.69/21.54% and 0/1.54%, respectively (App. Table 13).

Scope. The 2Wiki and RULER rows are disjoint from training but share its task families, so they measure task-adapted length transfer; the 8B deletion study and RULER continuation are separate protocols.

4.4 ATTRIBUTING THE SCHEDULE AND OPERATING RULE

The rule selects a basin, not a point optimum. Across 21 configurations spanning both a 99-run sweep and the exact-range factorial, the empirically selected τ never leaves a $0.75\times-1.5\times$ window around the rule value; an independent sweep selects $\tau=5$ with the rule value 5.657 only 0.0119 NLL away. Inside that window the best point moves with the configuration: over the 12 factorial configurations the rule is best in 4, $1.25\times$ in 6 and $0.75\times$ in 2, and the two pre-specified boundary arms land at $1.5\times$ (formula $\tau=1$) and $0.75\times$ (formula $\tau=8$). Thus $\tau=d_{\text{eff}}/\sqrt{L_{\text{train}}}$ is a search-free basin selector whose finite coefficient is empirical.

Cosh is a derived instance, not the only effective shape. The *matched exponential* is the arm $\rho \propto e^{-\tau'\phi}$ whose parameter is fixed so that its L^1 deformation from the uniform density equals that of rule-cosh, so the two arms move the same amount of budget with different curvature. It improves over Geo in 9/12 configurations and is not separated from rule-cosh: cosh-minus-exponential is $+0.00074$ NLL, bootstrap $[-0.0055, 0.0075]$, exact sign-flip $p=0.836$. The empirical result identifies interior allocation. Cosh contributes a closed-form, zero-parameter solution of the stated convex surrogate, not a uniqueness claim for trained objectives. The learned-frequency comparator and its tuning budget appear in App. E; shape attribution rests on the fixed schedules above.

The substrate persists under range scaling and channel scarcity. Two composition checks close the section. A fixed range operator has substrate-dependent leverage: on a 454M model ($L_{\text{train}}=2048$, 3 seeds), EVQ-COSH + RAMP reaches $100\pm 0\%$ teacher-forced NLL-gap retrieval at 8K against $61\pm 3\%$ for Geo + RAMP, and Karypis et al. (2026) independently report the same qualitative interaction for a learned non-geometric table. Under channel scarcity, a 432M MLA with only 16 rotary pairs improves $2\times$ extrapolation PPL from 138.8 to 95.6 (-31.1%) at $+0.9\%$ in-window cost across three seeds. Both are leverage results for the tested fixed operators, not claims over tuned range methods (App. G.2, G.1).

5 DISCUSSION

RoPE assigns a finite number of rotary channels to a continuous spectral range, and Theorem 1 says what that costs: usable positional dimension is the nominal $2K$ deflated by the mean redundancy $\bar{c}$. The exact-range experiment shows the deflation is not fixed by the operator, its base, or its sampled endpoints — moving only the interior channels changes the trained model. Frequency allocation is a design decision, not an implementation detail, and EVQ-COSH is one closed-form, zero-parameter way to exercise it.

Allocation and range transport act at different times on different objects: allocation shapes the table a model is trained or adapted on, transport acts on that table at inference. The RAMP and MLA results show the substrate still matters after a range operator is applied and when few rotary pairs exist, and the phase-matched OLMo experiment separates allocation from exposure — both arms receive identical target-8K/16K positions, data, updates, and trainable modules, yet Native reaches 2.02% RULER at 8K against 31.63%. The near tie at 4K on 2Wiki exact (22.0/21.5%) shows the in-window trade is task-dependent rather than a uniform loss of capability.

The budget accounting of §3.5 makes that trade falsifiable: thinning the intermediate $L < \lambda \leq M$ band should cost more on simultaneous multi-reference resolution than on single-reference retrieval, and a post-hoc split of the named NIAH families agrees in direction (-21.7 against -52.3 points at 4K). What the theory supplies is a budget identity, its slow-end collapse, and an exact obstruction to repairing a table swap after training; what it does not supply is a preferred curve or a certified operating point. That division is the durable part: operator design, spectral allocation, and deployment-time transport are distinct, composable parts of a positional-encoding system. A RoPE table spends a finite spectral budget, and where its interior channels land should be chosen deliberately, before that budget is transported.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. It changes only the initialisation of the RoPE inverse-frequency table and adds no learned parameters, so it

is inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

EVQ-COSH is fully specified by Eq. (6) and Eq. (7): the schedule is a pure function of (τ, K) that replaces the inverse-frequency buffer, with no learned parameters, auxiliary loss, or search, so any RoPE implementation can reproduce the intervention in a few lines without access to our training code. Complete proofs of Theorems 1, 3 and 4, Proposition 2, Corollary 5 and Lemma 6, together with every stated assumption, are in App. A. The identification protocol, including how sampled extrema and log-span are pinned, the exact pre-registered factorial design, and the corpora used by each arm, is in App. C; the mature-scale adaptation protocols, module budgets, phase-exposure mixtures, and evaluation harness settings are in App. F; remaining architectures and hyperparameters are in App. H. Seed scope is stated with every result.

AI USE STATEMENT

We used generative AI tools in this work, and we take full responsibility for the final content, including all text, claims, proofs, code, and experimental artefacts.

Within the categories for which ICLR requires disclosure, we used generative AI tools to help develop conceptual framings, to propose and refine hypotheses, to assist in drafting and checking proofs, to give feedback on experimental design, and to implement and interpret analysis code. We did not use generative AI tools to generate synthetic datasets used for training or evaluation, and we did not use them to produce any reported experimental number: every quantity in this paper is the output of an executed run, recorded with its protocol and seed scope. Within the categories for which disclosure is recommended, we also used these tools to draft and edit parts of the manuscript, to help produce figures, and to search and summarise related literature.

All AI-assisted work was reviewed by the authors. Each theorem and lemma stated here was re-derived and checked by an author, and the analytic identities were additionally verified numerically; AI-written code was read, tested, and its outputs cross-checked against independently recorded run artefacts; every citation was manually verified against the cited source; and all AI-suggested framings were checked against the underlying evidence before being written into a claim.

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A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_*(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_*(s) - b_*(d) \\ b_*(s) + b_*(d) & a(d) - a(s) \end{bmatrix}, \quad (8)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Proof of Theorem 1. R is the block matrix with blocks $R_{ij} = Q_{\omega_i \omega_j}$ and $R_{ii} = I_2$. Hence $\text{tr } R = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(R^2) = \sum_{i,j} \text{tr}(R_{ij} R_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $R_{ji} = R_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (3). $\square$

The identity is exact and involves only the mean of the pairwise redundancies; it does not claim that $\bar{c}$ determines the Shannon effective rank or the log-determinant, which retain higher-order multi-subspace dependence.

Proof of Proposition 2. With $t = \Delta/L$ and $x = \omega L$, $\cos(xt) = 1 - x^2 t^2/2 + O(x^4)$ and $\sin(xt)/x = t - x^2 t^3/6 + O(x^4)$, so after rescaling the second coordinate the basis of V_ω tends to $\{1, \Delta\}$. Substituting these expansions into (8) and whitening, the leading deficit between two slow subspaces is $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$; numerically at $x=0.05, y=0.10$ the exact-to-leading ratio is 1.00058. For the softmax metric, $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so constants are annihilated; centring by $\mathbb{E}_p$ and rescaling gives $\sin(\omega\Delta)/\omega \rightarrow \Delta - \mathbb{E}_p \Delta$ and $-2\overline{\cos}(\omega\Delta)/\omega^2 \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2$, which span the centred limit whenever p has nondegenerate support on at least three distances. $\square$

A.2 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 3. Both R_Ω and $R_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $R_\Omega(0) = R_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top R_{\Omega'}(\Delta) A^{-\top} = R_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $R_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation; the remaining freedom is a change of basis inside each two-dimensional block. For integer positions the same argument applies modulo the 2π alias. $\square$

This is an exact statement about *fixed, position-independent, invertible* compensation. It does not bound approximate compensation obtained by retraining on finite data, nor does it constrain nonlinear maps or the V/O and residual paths.

A.3 COLLISION-ONLY SURROGATE STATIONARY DENSITY

This subsection proves Theorem 4 directly from the collision-only surrogate functional; the forced Fisher extension is treated separately in §A.4. Starting from the formal surrogate kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (9)$$

the main-text collision functional is (4), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha\rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (10)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha\|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (11)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (12)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (13)$$

since differentiating $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha\rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (12) and (13) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (14)$$

which is main-text Theorem 4. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_\tau(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (14).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (15)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

A.4 FORCED FISHER EXTENSION

Adding a Fisher utility $-\mu \int \rho(\phi) b^{-2\phi} d\phi$ to $\mathcal{C}_{\text{app}}$ modifies the constrained stationarity to $\alpha\rho + \beta g - \mu b^{-2\phi} + \nu = 0$, whose twice-differentiated form is

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \quad (16)$$

with γ set by the Fisher multiplier’s sign convention. The non-resonant general solution ($\tau \neq 2 \log b$) is

$$\rho_{\text{full}}(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + \frac{\gamma}{(2 \log b)^2 - \tau^2} b^{-2\phi}. \quad (17)$$

At resonance, $\tau = 2 \log b$, the forcing profile $b^{-2\phi}$ coincides with one homogeneous mode. The particular solution is then

$$\delta\rho_F^{\text{part}}(\phi) = -\frac{\gamma}{2\tau} \phi e^{-\tau\phi}, \quad (18)$$

with the homogeneous constants again fixed by the boundary and mass conditions. All bounds below containing $(s^2 - \tau^2)^{-1}$ are non-resonant bounds and should be read with $|s^2 - \tau^2|^{-1}$; they are not uniform as $\tau \rightarrow 2 \log b$. EVQ-Cosh is the $\gamma=0$ homogeneous branch. The forced term is a controlled but nonzero residual; its magnitude and the corresponding warp-amplification caveat are bounded in §A.18.

A.5 CLOSED-FORM SOLUTION AND GEOMETRIC LIMIT

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + P b^{-2\phi}. \quad (19)$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2 \rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper’s “tether + pulse” interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^\phi \rho^*(s) ds, \quad (20)$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau). \quad (21)$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \rightarrow 0$. Then

$$\phi_k(\tau) \rightarrow 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \rightarrow u_k, \quad (22)$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.6 WHY THE PURE-TETHER BRANCH

The full stationary density contains three terms: a hyperbolic tether ($\cosh, \sinh$) and a particular solution $P b^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) Conditional CDF control. Under the slowly varying Fisher-coefficient assumption in Appendix A.18, the forcing profile is confined to a boundary layer: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$. This yields an $O(1/\log b)$ L^1 /CDF residual, not a pointwise bound. Inverse-CDF inversion can further amplify that residual by $1/\inf \rho_\tau \sim \sinh \tau/\tau$. Thus the pure-tether branch is adopted for its closed form; the residual analysis states when that choice is controlled and does not identify a trained-loss optimum.

A.7 WHY CONSTANT α (AND NOT THE CONTINUUM STATIONARY-PHASE COEFFICIENT)

A stationary-phase treatment of the exact kernel gives a ϕ -dependent diagonal coefficient rather than the constant coefficient used in $\mathcal{C}_{\text{app}}$. With $\lambda_b := \ln b$, its leading form is $\alpha_{\text{sp}}(\phi) = (\pi D_0 / L \lambda_b) b^\phi$. Writing the finite-grid coefficient as $\alpha_K := 1/(2K)$, the discrepancy is

$$\int_0^1 \rho^2(\phi) [\alpha_K - \alpha_{\text{sp}}(\phi)] d\phi = \frac{\|\rho\|_2^2}{2K} - \frac{\pi D_0}{L \lambda_b} \int_0^1 \rho^2(\phi) b^\phi d\phi, \quad (23)$$

which is not a higher-order correction at the deployed finite grids. Replacing α by $\alpha_{\text{sp}}(\phi)$ produces a different variable-coefficient Euler–Lagrange equation. We use constant α because it is the finite-grid surrogate that yields a positive density with an elementary inverse CDF; this is a modelling decision, not a continuum asymptotic claim. Its empirical check is therefore functional: whether the resulting schedule also reduces the separately evaluated exact-kernel collision score.

A.8 SURROGATE QUALITY: FUNCTIONAL VALIDATION

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the primary approximation behind the closed-form density; selecting the pure-tether branch and a finite operating point are additional modeling choices. We test the surrogate functionally rather than pointwise against the oscillatory exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This checks the direction selected by the homogeneous branch; it does not bound the omitted Fisher-forcing residual.

Table 5 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). These diagnostics use the inclusive grid $u_k = k/(K-1)$; they test the allocation direction but are not a direct validation of the deployed midpoint grid. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$; this is not a Cartesian sweep over every listed factor.

Table 5: Functional surrogate check on the inclusive $u_k = k/(K-1)$ diagnostic grid. EVQ reduces collision and increases effective rank under the *exact* kernel K in all 12 tested configurations. This is a direction check, not a pointwise approximation or a deployed-grid identity test.

Configuration	Collision C		Eff. rank	
	GEO	EVQ (Δ_C)	GEO	EVQ
<i>Text: $K=32$, $b=500K$, sweep L</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32$, $L=2048$, sweep b</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16$, $L=32$, sweep b</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ reduces collision under the exact kernel in all 12 configurations (24%–92%), and effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.9 EXACT HIGHER-ORDER DISPLACEMENT FLOOR

The one-channel mid-displacement threshold admits an exact Taylor series in $1/K$ that is fully closed-form (no empirical calibration).

Proposition 7 (Higher-order floor). *For the EVQ-Cosh mapping with $K = d_{\text{rot}}/2$ channels, as $N/K \rightarrow 0$ the N -channel mid-displacement threshold satisfies*

$$\tau_{\text{floor}}(N, K) = 4\sqrt{\frac{N}{K}} \left[1 + \frac{N}{2K} + \frac{241}{120} \left(\frac{N}{K} \right)^2 + O((N/K)^3) \right]. \quad (24)$$

All coefficients are derived from the Taylor series $\Delta\varphi(1/2, \tau) = -\tau^2/16 + \tau^4/256 + 17\tau^6/30720 + O(\tau^8)$, whose coefficients $(-1/16, +1/256, +17/30720)$ are exact rationals from the expansion of $\text{arcsinh}(\sinh \tau/2)/\tau$ (Appendix script `theory_B_floor_higher_order.py`; verified symbolically).

Accuracy. For $N=1$, against the exact floor obtained by solving $|\Delta\varphi(1/2, \tau)| = 1/K$, the leading term has relative error 3.85% at $K=16$ and 0.10% at $K=512$; including both displayed correction terms reduces the error to 0.088% at $K=16$, $1.56 \times 10^{-4}\%$ at $K=128$, and $2.41 \times 10^{-6}\%$ at $K=512$. All constants come from the cosh geometry.

A.10 SINGLE-CROSSING BUDGET SHIFT

Proof of Lemma 6. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi))/\sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau/\sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (??). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n(2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\text{arcosh}(\sinh \tau/\tau) \geq \tau/\sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1 - \phi))/\sinh \tau$ at $\phi_L = \log(L/2\pi)/\log b$ and $\phi_M = \log(M/2\pi)/\log b$. $\square$

A.11 WATERBED ENVELOPE AND COROLLARY

Setup. With $\rho > 0$, $\int_0^1 \rho = 1$, $U(d\phi) = d\phi$, $P_\rho(d\phi) = \rho d\phi$, $\delta\rho = \rho - 1$, and $M_\rho(s) = \int_s^1 \delta\rho du$, define the waterbed functional

$$\mathcal{W}(\rho) := \alpha \chi^2(U\|P_\rho) + \beta \int_0^1 M_\rho(s)^2 ds, \quad \chi^2(U\|P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho} d\phi. \quad (25)$$

The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the integrated squared cumulative error, which arises from the min-kernel off-diagonal covariance in K_{app} .

Proposition 8 (Waterbed inequality). *Let $D_B(\rho) = -\int_0^1 \log \rho(\phi) d\phi$ be the Burg entropy deficit. Then*

$$\mathcal{W}(\rho) \geq \alpha F_*(D_B(\rho)), \quad F_*(d) = \inf_{\rho > 0, \int \rho = 1, -\int \log \rho = d} \left[\int_0^1 \frac{1}{\rho} - 1 \right], \quad (26)$$

with equality $\mathcal{W}(\rho) = 0$ if and only if $\rho \equiv 1$ (geometric RoPE). A readable corollary via Rényi monotonicity is

$$\mathcal{W}(\rho) \geq \alpha(e^{D_B(\rho)} - 1). \quad (27)$$

Proof sketch. Since $\beta \int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha \chi^2(U\|P_\rho)$. For fixed $D_B(\rho) = d$, the sharp lower envelope of χ^2 over all densities is the Csiszár–Vajda f -divergence comparison F_* , whose extremizer is a two-level density $X = 1/\rho \in \{a, c\}$ with mass-normalization and entropy-deficit constraints. For the corollary, $1 + \chi^2(U\|P_\rho) = \int 1/\rho \mathbb{E}_U[\cdot] \geq e^{D_1(U\|P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality at $\rho \equiv 1$ is immediate from both sides vanishing. $\square$

EVQ evaluation. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ (Eq. (29) below uses the un-normalized integral; cf. §A.19 “Normalization convention”),

$$D_B(\rho_\tau) = \log \frac{\sinh \tau}{\tau} - \frac{1}{\tau} \int_0^\tau \log \cosh y \, dy = \frac{\tau^4}{90} - \frac{8\tau^6}{2835} + O(\tau^8), \quad (28)$$

$$S_{\chi^2}(\rho_\tau) = \frac{\sinh \tau}{\tau^2} \arctan(\sinh \tau) - 1 = \frac{\tau^4}{45} - \frac{2\tau^6}{315} + O(\tau^8), \quad (29)$$

$$\int_0^1 M_{\rho_\tau}(s)^2 \, ds = \frac{2\tau^4}{945} + O(\tau^6). \quad (30)$$

Combining,

$$\mathcal{W}(\rho_\tau) = \left(\frac{\alpha}{45} + \frac{2\beta}{945} \right) \tau^4 + O(\tau^6), \quad \mathcal{W}(\rho_{\tau_*}) = O\left(\frac{d_{\text{head}}^4}{(L_{\text{eff}}^J)^2} \right). \quad (31)$$

The centered companion functional is $O(\tau^4)$. It is not the value difference of $\mathcal{C}_{\text{app}}$ itself and does not determine trained-model NLL.

A.12 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 \, d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) \, d\phi \, d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ is the normalized cosh density.

Theorem 9 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (32)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) \, d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (33)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) \, d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (34)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (32). $\square$

An immediate corollary follows.

Corollary 10 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (35)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity is an exact internal check on the two surrogate energy terms; it makes no trained-loss claim.

A.13 FROM SCALING STRUCTURE TO AN OPERATING DEFAULT

The post-softmax model uses $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ with $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{head}}) + O(\tau^6)$ and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$. Its leading stationary balance is

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}. \quad (36)$$

Thus the model supplies the powers of d_{head} and L ; the finite coefficient $c(\Pi) = \sqrt{45\lambda Q_1}$ depends on the chosen utility, base, and protocol. Setting $c=1$ is the deployed zero-search convention, not an independently derived constant.

Here

$$q(x) = \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x}\right)^2 = \text{Var}_{t \sim U[0,1]}[\cos(xt)]$$

and $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$. This is a specified phase-variance utility. It should not be identified with the exact oscillatory collision objective or the trained-transformer loss.

The finite- τ evidence is deliberately empirical. A staged 99-run study contains nine small-model configurations and 27 formula-versus-Geo seed pairs, not 27 independent configurations. The default beats midpoint-Geo in 7/9 configuration means and 18/27 pairs, but beats a pilot-selected neighbouring value in only 3/9 held-out configuration means. In the separate exact-range factorial, the formula point is the best pre-specified Cosh multiplier in 4/12 structural configurations. These audits establish the correct role of $c=1$: a useful, fallible basin initialisation.

A.14 SCALING ANALYSIS OF τ^* : FROM SURROGATE TO OPERATING POINT

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha \delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L} \left[\frac{\sin((\omega_1 - \omega_2)L)}{\omega_1 - \omega_2} + \frac{\sin((\omega_1 + \omega_2)L)}{\omega_1 + \omega_2} \right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \dots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \quad \beta \sim L^{-0.22}, O(1). \quad (37)$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L . The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

The two parameters called τ . Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \quad (38)$$

This fitted-ratio scaling is neither the deployed rule nor evidence for it. The broadband surrogate selects the Cosh family; the separate post-softmax model in Proposition 11 balances an $O(\tau^2)$ utility with an $O(\tau^4)$ stiffness and yields

$$\tau_*^2 \propto \frac{d_{\text{eff}}^2}{L}. \quad (39)$$

The equality $c=1$ in the deployed rule is then evaluated by trained-model controls as described in §A.13; it is not obtained from the fitted β/α ratio.

A.15 POST-SOFTMAX OPERATING-POINT MODEL

The deployed scale is selected by a model separate from the surrogate that selects the shape. We state it here rather than in the main text because its small- τ expansion does not certify the deployed finite τ .

Proposition 11 (Semi-analytic operating-point scaling). *On the pure-tether family, assuming $Q_1(L, b) > 0$ and a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load stiffness and the post-softmax transport utility admit small- τ expansions $S_{\chi^2}(\tau) = c_S \tau^4/d_{\text{head}} +$*

$O(\tau^6)$ with $c_S = \frac{1}{45}$, and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$, where $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$ and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$. The leading-order stationary point of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ satisfies

$$\tau_*^2 = 45\lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

The explicit $d_{\text{head}}/\sqrt{L}$ factor is modulated by $Q_1(L, b)$, so this is a scaling structure rather than a pure power law, and the prefactor $c(\Pi) = \sqrt{45\lambda Q_1(L, b)}$ remains convention- and protocol-dependent. Two caveats bound what the proposition supports. The expansion is a small- τ statement, whereas the deployed configurations use τ near 4; at those values the exact cosh warp is evaluated directly and the series only motivates the functional form. The diffuse baseline $p_0 = 1/L$ is also a modelling assumption: trained attention is neither uniform nor stationary across heads, and the softmax-Jacobian effective window L_{eff}^J of §A.17 is the object that would replace it, but we do not measure it here. The deployed $c=1$ is therefore an empirical operating default, evaluated by the trained-model controls of §4.4, not a consequence of this proposition.

A.16 STIFFNESS FUNCTIONAL DERIVATION

The softmax transport objective in Proposition 11 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int_0^1 (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$.

Table 6: L -exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int (\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Exponent-matched diagnostic	0.80	0.498	Sensitivity analysis, not used in deployment
Pearson χ^2	1.0	0.465	Channel-load variance penalty (axioms)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi ($\alpha=2$)	—	0.807	

Why χ^2 . Two independent arguments favor $p=1$. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int (\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ’s non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2 .

Closed form. For $p=1$, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \cdot \arctan(\sinh \tau)}{\tau^2} - 1 \right], \quad (40)$$

verified numerically to relative error $< 7 \times 10^{-6}$.

Exponent-matched diagnostic. As a sensitivity analysis, one can ask which member of the S_p family would match the target exponent $\tau_* \propto 1/\sqrt{L}$ exactly over this grid. The resulting $S_{\text{derived}}(\tau)$ lies near $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), close to but distinct from the Pearson χ^2 choice $p=1$. We do not use this as a derivation of the stiffness; it is reported as sensitivity analysis. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ~60s).

A.17 EFFECTIVE WINDOW L_{eff}^J VIA SOFTMAX-JACOBIAN RAYLEIGH QUOTIENT

From density perturbation to RoPE frequency shift. The EVQ pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1} \operatorname{arsinh}((1-u) \sinh \tau)$. Setting $\theta = \tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta a(\phi) + O(\theta^2)$ with $a(\phi) = (1-\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) = u + \tau^2 h(u) + O(\tau^4), \quad h(u) = -\int_0^u a(v) dv = -\frac{1}{6}u(1-u)(2-u), \quad (41)$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)}[1 - \tau^2(\log b)h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)} = b^{-u_k}$. Hence $\dot{\omega}_k := \partial_\theta \omega_k|_{\tau=0} = -(\log b)\omega_k^{(0)}h(u_k)$.

Explicit RoPE logit transport direction. Fix a layer/head and query i ; write each 2D RoPE pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1} + iq_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1} + ik_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}} \overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} + iD_{ij,k}$. With relative position r_{ij} and attention scale s_{att} , the RoPE logit is $z_{ij}^{\text{rope}}(\tau) = s_{\text{att}}^{-1} \operatorname{Re} \sum_k \alpha_{ij,k} \exp(ir_{ij}\omega_k(\tau))$. The leading logit transport direction $g_i(j) := \partial_\theta z_{ij}^{\text{rope}}|_{\tau=0}$ has the closed form

$$g_i(j) = \frac{(\log b) r_{ij}}{s_{\text{att}}} \sum_{k=1}^K \omega_k^{(0)} h(u_k) \left[C_{ij,k} \sin(r_{ij}\omega_k^{(0)}) + D_{ij,k} \cos(r_{ij}\omega_k^{(0)}) \right]. \quad (42)$$

This is the Fréchet derivative of the logit with respect to the EVQ allocation perturbation at fixed activations. A finite- τ implementation that avoids first-order phase expansion at large $r_{ij}\omega$ is

$$g_i^{(\tau)}(j) = \frac{1}{s_{\text{att}}\tau^2} \operatorname{Re} \sum_{k=1}^K \alpha_{ij,k} \left[\exp(ir_{ij}b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij}b^{-u_k}) \right], \quad (43)$$

with $g_i(j) = \lim_{\tau \rightarrow 0} g_i^{(\tau)}(j)$.

Primary definition: Rayleigh quotient of the softmax Jacobian. For the trained attention distribution $p_i(j)$ on keys $j \in \mathcal{K}_i$, $J_{\text{sm}}(p_i) = \operatorname{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\text{sm}}(p_i) g_i = \operatorname{Var}_{j \sim p_i}[g_i(j)] = \frac{1}{2} \sum_{j,k} p_i(j) p_i(k) (g_i(j) - g_i(k))^2$. Since $J_{\text{sm}}(p_i) \mathbf{1} = 0$, we center g_i by the key-wise mean operator P_i . The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\text{eff}}^J} = \kappa_{\text{att}} = \frac{\mathbb{E}_{\ell,h,i}[g_i^\top J_{\text{sm}}(p_i) g_i]}{\mathbb{E}_{\ell,h,i}[\|P_i g_i\|_2^2]}, \quad (44)$$

This is the directional curvature of the *schedule-change* KL: for $p_\epsilon = \operatorname{softmax}(z + \epsilon g)$, $D_{\text{KL}}(p||p_\epsilon) = \frac{1}{2} \epsilon^2 g^\top J_{\text{sm}}(p) g + O(\epsilon^3)$, and for diffuse $p=1/L$ the quotient reduces to $1/L$. The phase-variance utility Q_1 in Proposition 11 is a distinct pairwise positional functional. Replacing its $1/L$ factor by $1/L_{\text{eff}}^J$ is therefore a possible modelling extension, not a consequence of the KL identity and not an empirical result used by the paper.

A.18 PURE-TETHER BRANCH: FISHER-FORCING COEFFICIENT AND RESIDUAL BOUND

Setup. The full stationary ODE under $\mathcal{C}_{\text{app}}$ plus a Fisher utility is $\rho'' - \tau^2 \rho = \gamma b^{-2\phi}$. Writing $s = 2 \log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta \rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha \rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi) b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) = \frac{1}{2s_{\text{att}}^2} \mathbb{E}_{\ell,h,i,j} [w_{\ell hij} |\alpha_{\ell hij,k}|^2], \quad \alpha_{\ell hij,k} = q_{i,k}^{\mathbb{C}} \overline{k_{j,k}^{\mathbb{C}}}. \quad (45)$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2 \rho = (\lambda_F/\alpha) V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\gamma(b) = \frac{\lambda_F}{\alpha} \bar{\eta}_F s^2 \left[1 + O\left(\frac{|\eta_F'|}{s \bar{\eta}_F} + \frac{|\eta_F''|}{s^2 \bar{\eta}_F}\right) \right], \quad s = 2 \log b. \quad (46)$$

The full $\eta_F(\phi)$ -local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F/\alpha)[\eta_F''(\phi) - 2s\eta_F'(\phi) + s^2\eta_F(\phi)]$.

Suppression is L^1 /CDF, not pointwise. The particular solution $\delta\rho_F^{\text{part}} = \gamma(b) q_s/(s^2 - \tau^2)$ combined with (46) gives pointwise amplitude

$$\delta\rho_F^{\text{part}}(\phi) \approx \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} e^{-s\phi}, \quad (47)$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta\rho_F^{\text{part}}$ is *not* pointwise small in s . The true suppression comes from the boundary-layer width of q_s : $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Let $C_{\text{bc}}^{(p)}(\tau)$ denote the finite L^p operator norm of the homogeneous correction enforcing the boundary and mass constraints, away from resonance. Then

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \leq C_{\text{bc}}^{(1)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \frac{1 - e^{-s}}{s} = O\left(\frac{1}{\log b}\right), \quad (48)$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \leq C_{\text{bc}}^{(2)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right), \quad (49)$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_\infty \leq \|\delta\rho_F\|_1 = O(1/\log b)$. For fixed Fisher coefficient and away from resonance, forcing is therefore small in CDF / L^1 transport at large b , but not necessarily pointwise small near $\phi=0$.

Warp amplification at large τ . Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_\infty \leq \frac{\|\delta\rho_F\|_1}{\inf_\phi \rho_{\text{full}}(\phi)}, \quad \inf_\phi \rho_{\text{EVQ}}(\phi) = \frac{\tau}{\sinh \tau}, \quad (50)$$

so, when the residual is small enough that ρ_{full} remains positive and comparable to ρ_{EVQ} , CDF perturbations can be amplified at the scale $\sinh \tau/\tau$. The bound does not justify omitting the forcing branch at large τ .

Dimensionless diagnostic. Using $\|\rho_{\text{EVQ}} - 1\|_2 = \tau^2/\sqrt{45} + O(\tau^4)$ and $\Gamma_b := (\lambda_F \bar{\eta}_F/\alpha) s^2/(|s^2 - \tau^2|)$, the L^2 -level ratio is

$$R_F := \frac{\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2}{\|\rho_{\text{EVQ}} - 1\|_2} \leq C_{\text{bc}}^{(2)} \Gamma_b \sqrt{45} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right). \quad (51)$$

Second-order utility deficit. Let A_τ be the constrained Hessian of the surrogate on zero-mass perturbations and let $q_s^\perp$ be the projection of q_s onto that tangent space. The full branch improves the quadratic objective by

$$\Delta_F = \frac{1}{2} \gamma(b)^2 \langle q_s^\perp, A_\tau^{-1} q_s^\perp \rangle, \quad (52)$$

so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \leq r$, the missed correction is $O(r^2)$ relative to the tether deformation energy.

A.19 CHANNEL-LOAD MOTIVATION FOR PEARSON χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1 - \rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4/(45 d_{\text{head}}) + O(\tau^6)$ consistent with Proposition 11. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha = 1/d_{\text{rot}} (= 1/d_{\text{head}}$ for MHA via §A.14; $= 1/d_{\text{rot}}$ for compressed-RoPE MLA via §G.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi) = 1/\rho(\phi)$ and samples channels according to $dP_\rho = \rho d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1 - \rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (53)$$

We treat this as a canonical motivated choice under the stated axioms, not a uniqueness claim.

Load-moment exponent. Distinct from the weighted- L^2 family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ used in Table 6 (where Pearson χ^2 is $p=1$), we also examine a load-moment family $S_p^{\text{mom}}(\rho) = \mathbb{E}_\rho[(1/\rho - 1)^p]$. Substituting the EVQ cosh density and minimizing $S_p^{\text{mom}} - \lambda U$ gives $\tau^{2(p-1)} \asymp d_{\text{head}}^2 / L_{\text{eff}}^J$, hence $\tau^* \propto (L_{\text{eff}}^J)^{-1/(2p-2)}$; only $p=2$ yields the observed $L^{-1/2}$ exponent within this family. The two families parametrize different f -divergence neighborhoods and need not match on the finite- τ grid.

f -divergences at finite τ . Smooth f -divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho=1$. Over the finite grid used in Table 6, however, their fitted exponents range from 0.58 to 0.81. This is model sensitivity, not a trained-PPL comparison.

A.20 DISCRETE-CHANNEL TRANSPORT GAP AND THE MLA CHANNEL-SCARCITY MECHANISM

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (54)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (55)$$

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$, $m_\tau = \tau / \sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (56)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km} + O(K^{-2}). \quad (57)$$

The singular δ -component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w ,

$$\mathcal{D}_K[\rho] \approx \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (58)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of the distortion bound by $4\times$ and the K^{-1} component by $2\times$. The observed MLA gain (§??) is a test of the predicted direction, not a controlled quantitative test of either bound.

Caveat. At large τ , $m_\tau = \tau / \sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds can therefore become vacuous in the same regime where the Fisher-residual analysis no longer guarantees a small warp error (§A.18).

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model primary protocols are documented separately in Apps. C and F. This section records the submitted scratch-model and cross-modal experiments that remain supporting evidence for the substrate-range and cross-modal scope claims. The learned-frequency comparator appears in App. E.

B.2 REPRODUCIBILITY SNAPSHOT

Table 7: Compact reproducibility summary for supporting scratch-model experiments.

Setting	Train length	Scale	Seeds	Main datasets
EVQ×RAMP	2048	454M	3 + 3	FineWeb-Edu + 10% passkey mix
MLA	8192	432M	3	FineWeb-Edu

Table 8 lists the training hyperparameters shared across the supporting text experiments.

Table 8: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 3×10^{-4} (125M), 2×10^{-4} (454M/MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K geometric checkpoint and applies a 500M-token continuation with either the geometric or EVQ table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. This is a single-seed post-hoc continuation result, not a from-scratch allocation control.

Table 9: Single-seed 750M continued-pretraining check (2K→ 4K, 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials. This supporting row is separate from the multi-seed primary controls.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$, whereas the best tested Video DiT point is $\tau=1.5$ ($0.53\times$). We use this empirical correction only to select the supporting video configuration; the sweep does not derive a modality-general coefficient.

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

We evaluate a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) with temporal RAMP. For each seed, the geometric and EVQ-Cosh ($\tau=1.5$) branches are trained in the same head-to-head run.

Table 10 reports two independent seeds. The within-seed contrasts favor EVQ on train, all-frame extrapolation, and far-frame extrapolation MSE in both runs; the two-seed mean deltas are -26% , -16% , and -33% . This is a cross-modal scope check, not an estimate of a general video effect size.

Table 10: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Each seed is one shared-run comparison; values are denoising MSE at $t=0.5$.

Seed	Metric (RAMP)	Geo ($\tau=0$)	EVQ ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	-21%
42	All extrap	0.00724	0.00606	-16%
42	Far extrap	0.00989	0.00639	-35%
137	Train MSE	0.01113	0.00761	-32%
137	All extrap	0.00713	0.00598	-16%
137	Far extrap	0.00856	0.00601	-30%
Mean Δ		$-26\% / -16\% / -33\%$		

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. In the pre-specified diagnostic, $\tau=1.2$ leaves 11/16 channels above this threshold (effective rank 9), whereas $\tau=1.5$ leaves 12/16 (rank 10). Repeating the head-to-head comparison at base=1,000 places 13/16 geometric channels above the threshold. In this single-seed run, $\tau=1.2$ and $\tau=1.5$ differ by less than 0.3% on the reported metrics and both improve far-frame MSE by about 48% relative to Geo. This is consistent with the diagnostic; it is not an equivalence or causal estimate.

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed mid-point grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$

Table 11: Base = 1,000 DiT head-to-head (129.6M, seed 42, RAMP). The two EVQ settings are within 0.3% on the reported metrics in this run.

τ	Train MSE	All extrap	Far extrap	$\Delta_{\text{far vs Geo}}$
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	0.00933	0.00820	0.00859	−48%
1.5	0.00930	0.00822	0.00858	−48%

has relative density 0.76. Thus thinning concentrates on channels with $1/\omega \gg L_{\text{train}}$ while preserving more of the training-scale band. This motivates a directional prediction: channels with $1/\omega \ll L_{\text{train}}$ complete many periods during training, so positions beyond L_{train} land in already-trained phase regions, while channels with $1/\omega \gg L_{\text{train}}$ never complete one and extrapolate into unseen phase. Moving budget toward the former should favour extrapolation, while any in-window cost must be measured after training. The surrogate does not assign the much larger mismatch observed when a pretrained model is retrofitted with a new table.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global and micro batch; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

Arms. *Baseline:* a paper-faithful reimplementation of the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026), at training base 256. We did not identify a public author implementation when implementing the control, so the comparison is against the rule as published, not an official-code reproduction. *Intervention:* endpoint-normalised cosh interior spacing at $\tau=4$. Note that the endpoint-normalised cosh arm is *not* identical to the deployed midpoint EVQ-COSH grid of Eq. (7), whose τ also shifts the sampled extrema; endpoint normalisation is what makes the exact-range control possible, and this is a shape diagnostic rather than a deployment configuration.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Seed-42 absolute values. Under *fixed* range, PPL is 28.386/142.851/342.056/564.353 (uniform) against 29.331/88.615/278.659/504.131 (cosh) at 256/512/1,024/2,048. Under *target-matched* range, 28.386/28.038/35.209/100.111 (uniform) against 29.331/29.804/42.230/132.271. Cosh wins 32/32, 27/32, 22/32 anchors at 512/1K/2K under *fixed*, and only 5/32, 1/32, 2/32 under *target-matched*.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. The *target-matched* column bounds the deployment claim: when range transport is available to both arms, target-aware retargeting is the stronger lever in this protocol. Both are reported in the body (§4.2).

C.2 EXACT-RANGE FACTORIAL AT 50.9M

Design. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected. Grid: 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 structural configurations;

5 schedules (Geo, anchored cosh at $0.75\times$, $1.00\times$ [the formula point], $1.25\times$, and a deformation-matched exponential); 3 seeds. Main factorial 180/180 completed, boundary- τ follow-up 12/12 completed, covering the complete pre-specified matrix.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); cosh $0.75\times$ 6.202858 (494.16); cosh $1.00\times$ 6.202094 (493.78); cosh $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. Optima move inward from both ends, which bounds the formula’s implied range but does not yield a replacement closed-form rule.

Scope limit. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated local WikiText-2 raw stream — not a large-scale pre-training distribution, and a different corpus from the 151.9M control and from the from-scratch runs. It is not scale evidence and not capability evidence, and its effect size is not comparable to the 151.9M single-configuration contrast: the factorial establishes cross-configuration *direction*, whereas the 151.9M control provides the effect size within its configuration.

D THE FIXED-INDEX RAMP SCALER

Every arm marked +RAMP in this paper uses our fixed-index smooth-ramp range scaler from the NTK-by-parts family that includes YaRN (Peng et al., 2024). It is used as a fixed-scale substrate control, not as a reproduction or tuning study of reference YaRN.

Its role is a fixed-scale substrate control: it is held *identical* across the arms being compared — same implementation, same fixed cut indices, same scale s — so that the only difference between Geo + RAMP and EVQ-COSH + RAMP is the training-time table each substrate presents to the same operator. It therefore measures substrate-dependent leverage at the fixed tested scale, rather than tuned range-method performance.

Implemented components. For K frequency channels, the scaler uses fixed channel-index cut points $k_{\text{lo}} = \lfloor 0.20K \rfloor$ and $k_{\text{hi}} = \lfloor 0.90K \rfloor$. Between them it applies the cubic smoothstep $r(x) = x^2(3 - 2x)$ and replaces each inverse frequency ω_k by

$$\omega_k / \left(s^{r_k} t^{r_k/2} \right), \quad t = 1 + 0.07 \log_2 s.$$

Thus an interpolation ramp is present, but its boundaries are fixed by channel index rather than YaRN’s wavelength-derived $\beta_{\text{fast}}/\beta_{\text{slow}}$ rule. The factor t is folded into frequency scaling; there is no separate attention-logit magnitude scaling. These differences are why we label the operator RAMP rather than YaRN.

E LEARNED-FREQUENCY COMPARATOR

The 32-parameter arm in Table 12 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at $10\times$ and $100\times$ the base rate, and the stronger $100\times$ setting is the 455.3 row. This answers a tuning-budget question but not the attribution question: a learned-parameter arm cannot isolate allocation shape from parameterisation, which is why §4.2 uses fixed

Table 12: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). **We rest no allocation-shape attribution on this table** (§4.4): the learned-inv-freq arm carries 32 trainable parameters, so any gap confounds allocation shape with parameterisation and optimisation effort. The arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026) rather than the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (layer-shared)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	− 35.0%

Table 13: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint but share the 13 generator families: task-adapted length transfer, not unseen-task transfer.

Model / protocol	Arm / metric	1 $\times$	2 $\times$	4 $\times$
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official macro	82.16	0.08	0
	EVQ-COSH, official macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official macro	72.19	2.02	0.38
	EVQ-COSH, official macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official macro	94.44	0.295	—
	EVQ-COSH, official macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

zero-parameter schedules under an identical operator instead, and why no allocation-shape claim in this paper depends on this table.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). The distinction matters for what the row can be read as evidence for. Read as a learned-table comparison, its direction is consistent with the independent finding that learned tables degrade more sharply than standard RoPE under naive extrapolation.

F MATURE-SCALE PROTOCOLS

F.1 OLMo-2 1.485B

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive end-point are reported separately in Table 4.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start independently from the matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH exact match is 22.0/21.5% at 4K, 0/17.5% at 8K, and 0/4.0% at 16K. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length. Thus this is held-out task-family length transfer, not an unmodified LongBench leaderboard result. The RULER adapters are independent of the QA adapters: Table 13 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families.

Family-level in-window structure. For the selective-Q/K RULER adapters at 4K, the official-score changes (EVQ-COSH minus Native) for `niah_single_1/2/3` are $-5/0/-60$ percentage points, averaging -21.67 . The changes for `niah_multikey_1/2/3`, `niah_multivalue`, and `niah_multiquery` are $-65/-95/-50/-26.25/-25$, averaging -52.25 . This split is a post-hoc mechanism diagnostic over one matched seed, not a separate benchmark claim; it motivates the multi-reference prediction in §5.

From initialisation. Starting from the public step-0 initialisation, one 2.097B-token EVQ-COSH trajectory reaches PPL 167.45/156.87/159.64 against the released Geo step-1,000 checkpoint’s 161.19/163.88/182.73 at 4/8/16K. The paired EVQ-COSH-minus-Geo NLL is $+0.0381/-0.0437/-0.1351$, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe trajectory: EVQ-COSH used a reviewed single-GPU HF loop, whereas the released Geo checkpoint used the original distributed trainer. This row evaluates natural-text LM; capability is evaluated by the separate autoregressive protocols above.

F.2 LLAMA-3-8B-INSTRUCT

Probability and causal source use. In the matched seed-42, 300-step Q/K/V/O LoRA protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Removing the remote gold block from every head changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and causal-routing measurements; autoregressive task accuracy is evaluated separately.

RULER-family continuation. A separate 516-step continuation uses identical physical-8K supervision over all 13 RULER families, with one seed per arm and disjoint generated training/evaluation rows. At 16K, Native/EVQ-COSH official macro is 0.295/14.03% and normalized exact is 0/1.54%. At the 8K training length the metrics disagree, so Table 13 reports both official macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, so they establish task-adapted length transfer, not unseen-task transfer. The natural-text and routing rows make no task-transfer claim.

G SUPPORTING RESULTS

G.1 MULTI-HEAD LATENT ATTENTION

We test the finite-budget hypothesis in a 432M-parameter MLA transformer with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens at $L_{\text{train}}=8192$ and base 500K over seeds

42, 43, and 88. EVQ uses the pre-specified $\tau=1.414$ setting. This experiment tests the frequency allocation at fixed architecture; it does not identify a general MLA operating rule for τ .

Table 14: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). The legacy scaler is this run’s wavelength-blend operator, not the RAMP used elsewhere.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
GEO+legacy(s=4)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ+legacy(s=4)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favor EVQ at 16K. EVQ also improves on GEO+legacy at 16K and ties it to printed precision at 24K; EVQ+legacy has the lowest mean PPL at every reported length. Because the legacy operator is not a matched-range control, these rows establish only the ordering in this architecture and protocol.

G.2 COMPOSITION WITH THE RAMP RANGE SCALER

H SUPPORTING EXPERIMENTS (FROM MAIN TEXT)

The following results record additional submitted protocols with their seed scope.

Cross-modal scope. In two 129.6M DiT shared-run comparisons, the EVQ branch has lower denoising MSE on train, all extrapolation frames, and far extrapolation frames in both seeds. Full values and the one-seed base diagnostic are in §B.5.

Table 15: **Substrate $\times$ range composition** (454M, $L_{\text{train}}=2048$, 10% passkey mix, 3 seeds). RAMP is our fixed-index smooth-ramp scaler (§4.1), applied at the *same* fixed scale $s=8$ to both substrates. PK is teacher-forced NLL-gap retrieval; PK@8K is mean $\pm$ std, other entries are 3-seed means.

Method	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41 $\pm$ 5%	57%	51%	161.9	253.2
Geo+RAMP	61 $\pm$ 3%	59%	51%	82.9	157.7
EVQ-COSH	53 $\pm$ 8%	63%	50%	150.3	229.5
EVQ-COSH+RAMP	100$\pm$0%	79%	68%	70.9	107.5

Table 16: $L=256$ RAMP composition study (125M, 3-seed mean PPL). Among the tested settings, $\tau=4.0$ has the lowest mean PPL at every reported extrapolation ratio.

Method ($L=256$)	$4\times$	$8\times$	$16\times$	$32\times$
Geo+RAMP	60.7	97.5	190.4	260.2
EVQ-COSH ($\tau=2$)+RAMP	51.6	66.7	107.9	141.7
EVQ-COSH ($\tau=4$)+RAMP	50.9	60.9	84.1	99.6